

NORTHFIELD CREAMERY INGREDIENTS, INC.

1425 Kettle Creek Road · Northfield, Wisconsin 53012 · USA
Tel (262) 555-0148 · Quality Assurance: qa@northfieldcreamery.example · FDA Food Facility Reg. 18902457364

Specification No. SPEC-NCI-2280
Revision 5
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Supersedes: Rev. 4, 03 August 2024
Next Scheduled Review: March 2029
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PRODUCT SPECIFICATION SHEET

Product Name	Cheddar Cheese Powder, Medium	Product Code	NCI-2280
Common / Usual Name	Cheese Powder	Product Form	Free-flowing spray-dried powder
Manufacturing Site	Northfield, Wisconsin, USA — Plant Code 55-0421	Process	Blended, pasteurised, spray dried
Country of Origin	United States of America	Shelf Life	12 months from manufacture
Certification	SQF Food Safety Code for Manufacturing, Edition 9 — Certificate SQF-114772. Kosher certified (Dairy, cRc). Manufactured under a HACCP plan and a Preventive Controls food safety plan per 21 CFR 117. Current certificates are issued as their own documents and are not part of this specification.		

This specification describes the medium cheddar cheese powder supplied by Northfield Creamery Ingredients under item NCI-2280. It is intended for use as a flavour and colour ingredient in dry seasoning blends, snack topical applications, sauce and soup bases, and extruded snack coatings. It is **not** intended for retail sale as a finished consumer product without further processing. All values stated in the specification tables below are **acceptance limits for the product**; results measured on an individual production lot are reported separately on the certificate of analysis issued with each shipment.

INGREDIENT STATEMENT

The following statement is the declaration for this ingredient and must be carried through to the finished-product label in descending order of predominance:

INGREDIENTS: WHEY, CHEDDAR CHEESE (PASTEURIZED MILK, CHEESE CULTURES, SALT, ENZYMES), BUTTERMILK, SALT, DISODIUM PHOSPHATE, LACTIC ACID, ANNATTO (COLOR). **CONTAINS:** MILK.

No artificial flavours, no added MSG, no hydrolysed vegetable protein, no preservatives. All ingredients are GRAS or are used in accordance with 21 CFR, and no ingredient is bioengineered as defined by 7 CFR Part 66.

ALLERGEN MATRIX — FDA "BIG 9"

Allergen	In product	Shared line	In facility
Milk	YES — ingredient	Yes	Yes
Eggs	No	No	No
Fish	No	No	No
Crustacean shellfish	No	No	No
Tree nuts	No	No	No
Peanuts	No	No	No
Wheat	No	No	No
Soybeans	No	No	Yes — see note
Sesame	No	No	No

Soy lecithin is handled in Room 4 of the Northfield facility on dedicated equipment that shares no air handling, dust collection or product contact surface with the NCI-2280 line. Allergen changeover is validated by ATP swab and by protein-specific lateral flow at a 5 ppm limit of detection. Gluten: this product is not made from a gluten-containing grain and tests < 20 ppm gluten by R5 ELISA.

NUTRITIONAL INFORMATION — PER 100 G

Nutrient	Per 100 g	Method / basis
Energy	497 kcal / 2,079 kJ	Calculated
Total Fat	33.0 g	AOAC 989.05
of which Saturated Fat	20.5 g	AOAC 996.06
of which Trans Fat	1.1 g	AOAC 996.06
Cholesterol	95 mg	AOAC 994.10
Sodium	3,100 mg	AOAC 985.35
Total Carbohydrate	24.0 g	By difference
of which Dietary Fibre	0 g	AOAC 991.43
of which Total Sugars	18.0 g	AOAC 982.14
incl. Lactose	17.2 g	AOAC 984.15
Protein (N × 6.38)	26.0 g	AOAC 991.20
Calcium	780 mg	AOAC 985.35
Moisture	3.5 g	AOAC 927.05
Ash	13.5 g	AOAC 930.30

Typical values, per 100 g of product as sold. Not a lot-specific analysis.

CHEMICAL & PHYSICAL SPECIFICATION

Attribute	Specification	Unit	Method
Moisture	3.5 max	%	AOAC 927.05
Milkfat	32.0 – 34.5	%	AOAC 989.05
Protein (N × 6.38)	24.0 min	%	AOAC 991.20
Salt (as NaCl)	6.5 – 8.5	%	AOAC 971.27
Ash	14.0 max	%	AOAC 930.30
pH (10% w/v solution, 20°C)	5.2 – 5.8	—	AOAC 981.12
Bulk density, loose	0.42 – 0.52	g/mL	NCI-QM-014
Particle size, through US #40	95 min	%	ASTM E11
Scorched particles	Disc B max	—	ADPI

MICROBIOLOGICAL LIMITS

Test	Specification	Unit	Method
Standard Plate Count	30,000 max	CFU/g	AOAC 990.12
Coliforms	10 max	CFU/g	AOAC 991.14
Escherichia coli	Negative in 1 g	—	AOAC 991.14
Yeast	50 max	CFU/g	AOAC 997.02
Mould	50 max	CFU/g	AOAC 997.02
Staph. aureus, coag. pos.	10 max	CFU/g	AOAC 2003.07
Salmonella	Negative in 375 g	—	AOAC 2016.01
Listeria monocytogenes	Negative in 25 g	—	AOAC 2013.10

Micro limits apply to the product as shipped. Environmental monitoring of the dry-blend zone is performed weekly under NCI-QM-031.

PACK SIZE & CASE CONFIGURATION

Primary pack	50 lb (22.68 kg) net — 5-ply multiwall kraft bag with blue polyethylene liner, heat sealed
Case GTIN-14	10812345600282
Consumer unit UPC-A	812345600285 — 2 lb (907 g) foil pouch, item NCI-2280-02
Pallet configuration	8 bags per layer × 5 layers = 40 bags per pallet (Ti 8 / Hi 5); pallet net weight 2,000 lb (907.2 kg)
Pallet	48 × 40 in. heat-treated wood, ISPM-15 stamped, stretch wrapped with a top slip sheet
Bulk alternative	2,000 lb (907 kg) corrugated tote with liner, item NCI-2280-T — minimum order 4 totes

SHELF LIFE & STORAGE

Shelf life	12 months from date of manufacture in unopened original packaging
Storage conditions	Store below 75°F (24°C) at 65% relative humidity or less, in a dry, odour-free warehouse, off the floor and out of direct sunlight
After opening	Reseal the liner and use within 30 days
Date coding	Julian code YYDDD followed by the line number, ink-jetted on the bag gusset
Transport	Dry van, food-grade, previous-load restricted; refrigeration not required

COUNTRY OF ORIGIN

Country of Origin: United States of America. The milk, whey and buttermilk used in this product are sourced exclusively from Grade A dairy farms in Wisconsin and Minnesota. Blending, pasteurisation, spray drying and packing are all performed at our Northfield, Wisconsin facility (Plant Code 55-0421). No component of this product is imported. Country of origin does not change from lot to lot; any change to this statement will be issued as a new revision of this specification.

SENSORY

Appearance	Free-flowing powder, free of lumps and foreign matter
Colour	Uniform orange to orange-yellow
Flavour & odour	Clean, characteristic medium cheddar; no rancid, scorched or off notes

Prepared by: M. Okonkwo, Quality Assurance Technologist · 12 March 2026

Approved by: Dana Whitcomb, Director of Quality Assurance · 12 March 2026

Revision history: Rev. 4 (03 Aug 2024) sesame added to the allergen matrix · Rev. 5 (12 Mar 2026) pallet configuration corrected to Ti 8 / Hi 5, nutritionals re-issued per 100 g. Every figure in the tables above is an **acceptance limit for the product**. Measured results for an individual production lot are reported on the certificate of analysis that accompanies each shipment; this specification is the document those results are judged against. Uncontrolled when printed.